

Hero Adjusted Plus–Minus in *Marvel Rivals*

Intention-to-treat estimates from 477,483 competitive matches

Season 19, PC ranked. Prepared 8 September 2026. Commit 31cfdff.

Specification. One observation per match; the outcome is victory for side 0. Each hero enters as a signed contrast (own-side weight less opposing-side weight) under *starting-lineup* attribution (W0), reduced onto a basis that sums to zero *within each role*, so every hero receives both a coefficient and a standard error. Controls: a separate intercept per map, the pre-match rank-score differential, composition-shape indicators, and 100 team-up contrasts. Estimated by unpenalised logit with heteroskedasticity-robust (HC1) standard errors.

The estimand is intention-to-treat. W0 is the hero each player *started* on, recovered from entry 1 of the API hero array, which is chronological by first appearance. It is the only attribution fixed before any outcome information exists. A player who starts on a hero and abandons it after thirty seconds still carries full weight, so these effects are diluted by non-compliance — but how often a hero gets abandoned is part of its value, not an error to remove.

Why not play-time weighting. Table 2 shows effect magnitude rising monotonically with how much post-outcome information the attribution uses. The best-*fitting* rule is the most contaminated: play-time weighting predicts better precisely because its regressors encode what happened. Its effects are roughly 2.5 times larger than W0’s while the ranking is substantially preserved.

Why estimates are within role. Cross-role comparisons are not a hero property but a composition property. The composition-shape block can reproduce the role-count differential *exactly*, so under a single global sum-to-zero constraint the hero and shape blocks are collinear in that direction and are separated only by the 0.16% of teams pooled into an “other” category. Constraining β to sum to zero within each role makes it orthogonal to every role indicator, so composition can only land in the shape block, where the full sample identifies it. Refitting under this constraint leaves the reported estimates numerically unchanged (Spearman 0.9999, mean absolute difference 0.010 pp), confirming that the earlier post-hoc projection was already removing exactly the ill-conditioned direction.

Table 1: Hero adjusted plus–minus, relative to an average hero of the same role

	APM	Std. error	95% CI	W1
<i>Panel A: Tank (15 heroes)</i>				
The Thing	5.46***	(0.204)	[5.06, 5.86]	7.21
Magneto	1.96***	(0.139)	[1.69, 2.24]	3.57
Angela	1.87***	(0.261)	[1.36, 2.38]	2.68
Peni Parker	1.51***	(0.328)	[0.86, 2.15]	7.41
Hulk	1.33***	(0.219)	[0.90, 1.76]	3.55
Thor	1.29***	(0.219)	[0.86, 1.72]	0.65
The Hood	0.45**	(0.172)	[0.11, 0.78]	6.52
Devil Dinosaur	-0.03	(0.231)	[-0.48, 0.42]	-2.24
Groot	-0.67**	(0.290)	[-1.23, -0.10]	-3.79
Venom	-0.93***	(0.219)	[-1.36, -0.51]	-1.35
Doctor Strange	-0.98***	(0.202)	[-1.37, -0.58]	-6.82
Captain America	-1.22***	(0.352)	[-1.91, -0.53]	-1.78
Emma Frost	-1.77***	(0.192)	[-2.15, -1.39]	-6.68
Rogue	-3.41***	(0.344)	[-4.09, -2.74]	-1.78
Deadpool (Vanguard)	-4.86***	(0.252)	[-5.36, -4.37]	-7.15

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Table 1, continued

	APM	Std. error	95% CI	W1
<i>Panel B: Damage (27 heroes)</i>				
Black Cat	4.85***	(0.260)	[4.34, 5.36]	11.62
Magik	3.47***	(0.219)	[3.04, 3.90]	14.57
Storm	3.09***	(0.305)	[2.49, 3.68]	10.27
Daredevil	2.92***	(0.217)	[2.50, 3.35]	8.47
Mister Fantastic	2.52***	(0.351)	[1.83, 3.21]	3.95
Iron Man	1.34***	(0.279)	[0.79, 1.89]	3.10
Iron Fist	1.32***	(0.224)	[0.88, 1.76]	4.52
Hela	1.14***	(0.199)	[0.75, 1.53]	1.22
Winter Soldier	1.00***	(0.141)	[0.73, 1.28]	0.17
Wolverine	0.82**	(0.317)	[0.20, 1.44]	0.21
Spider-man	0.58***	(0.175)	[0.24, 0.93]	2.90
Moon Knight	0.10	(0.297)	[-0.49, 0.68]	-3.37
Black Widow	0.02	(0.200)	[-0.37, 0.42]	-1.24
Blade	-0.20	(0.317)	[-0.82, 0.43]	1.09
Black Panther	-0.21	(0.239)	[-0.68, 0.26]	2.87
Human Torch	-0.28	(0.351)	[-0.97, 0.41]	4.56
Elsa Bloodstone	-0.39	(0.229)	[-0.83, 0.06]	-0.12
Psylocke	-0.58	(0.342)	[-1.25, 0.09]	1.54
Deadpool (Duelist)	-0.67***	(0.220)	[-1.10, -0.24]	1.85
Namor	-0.82***	(0.243)	[-1.30, -0.34]	-7.57
Hawkeye	-1.22***	(0.341)	[-1.89, -0.55]	-2.25
The Punisher	-1.81***	(0.262)	[-2.33, -1.30]	-10.45
Star-lord	-2.41***	(0.300)	[-3.00, -1.82]	-4.05
Scarlet Witch	-2.49***	(0.414)	[-3.30, -1.68]	-5.91
Squirrel Girl	-3.78***	(0.350)	[-4.46, -3.09]	-16.61
Cyclops	-4.04***	(0.195)	[-4.42, -3.66]	-7.65
Phoenix	-4.28***	(0.345)	[-4.96, -3.61]	-13.67
<i>Panel C: Support (13 heroes)</i>				
Mantis	6.76***	(0.181)	[6.40, 7.11]	13.50
Rocket Raccoon	4.14***	(0.150)	[3.85, 4.44]	5.18
Ultron	3.59***	(0.297)	[3.01, 4.17]	6.76
Adam Warlock	2.08***	(0.315)	[1.47, 2.70]	3.93
Loki	1.18***	(0.268)	[0.65, 1.70]	1.84
Cloak & Dagger	-0.04	(0.170)	[-0.37, 0.29]	0.38
Gambit	-0.30	(0.210)	[-0.71, 0.11]	-0.64
Invisible Woman	-1.38***	(0.151)	[-1.67, -1.08]	-5.39
Jeff The Land Shark	-1.75***	(0.233)	[-2.21, -1.29]	-5.01
Jubilee	-1.91***	(0.172)	[-2.25, -1.58]	0.32
Luna Snow	-3.01***	(0.167)	[-3.33, -2.68]	-7.25
White Fox	-3.48***	(0.208)	[-3.89, -3.07]	-6.69
Deadpool (Strategist)	-5.89***	(0.210)	[-6.30, -5.48]	-6.91
Matches	477,483			
Player-matches	3,812,088			
Heroes	55			
Log-likelihood	-320,067.7			

Table 2: Effect magnitude rises with post-outcome contamination

Rule	Attribution	Log-lik.	Effect s.d.	vs W0
W0	starting lineup (pre-outcome)	−320,122	2.60	1.00×
W2	dominant hero	−305,869	4.27	1.64×
W1	play-time weighted	−297,648	6.38	2.45×

All three fitted on the same 477,483 matches. Effect s.d. is the standard deviation of within-role estimates in percentage points. Rank agreement with W0: $\rho = 0.937$ (W2), $\rho = 0.891$ (W1). Squirrel Girl reads −3.76 under W0, −8.62 under W2 and −16.61 under W1. Higher log-likelihood here indicates worse identification, not a better model.

Table 3: Raw win rate by number of Tanks fielded

Tanks	Teams	Win rate (%)	vs. two tanks
0	1,990	18.49	-33.81
1	89,534	39.44	-12.86
2	533,702	52.30	—
3	24,456	41.28	-11.02
4	288	34.72	-17.58

Composition from each player’s play-time-dominant hero. The relationship is an inverted U peaking at the 2–2–2 standard that 76% of teams field, so more Tanks is not better. A term linear in tank counts cannot represent this shape; because teams fielding too few Tanks outnumber those fielding too many by roughly four to one, the fitted slope is dragged positive. This is the artifact the within-role normalisation in Table 1 removes.

Notes. Units are percentage points of win probability evaluated at a balanced match ($\partial p / \partial x = \beta / 4$). Standard errors in parentheses are HC1 and are propagated through the within-role contrast by the delta method. Stars denote Benjamini–Hochberg false-discovery-rate control across the 55-hero family: * $q < 0.10$, ** $q < 0.05$, *** $q < 0.01$; 10 of the 55 heroes are not distinguishable from their role average. The final column repeats the estimate under play-time attribution (W1) rather than starting-lineup attribution (W0), on the *same* 477,483-match sample. Sample: 482,997 matches crawled, less 2,139 containing draws (`is_win` = 2), 19 with incomplete play time, and 3,356 below a 240-second forfeit floor. Four team-up contrasts were structurally empty and dropped.

Table 4: Specification checks

Check	Result	Interpretation
Placebo (permuted heroes)	0.023 vs 0.881	97.4% of signal destroyed
Sum-to-zero normalisation	-1.1×10^{-16}	holds at machine precision
Rank agreement, raw win rates	$\rho = 0.949$	up from 0.703 before correction
Attribution robustness (W1 vs W2)	$\rho = 0.954$	ranking stable; magnitudes are not

The placebo shuffles hero assignment against outcomes and refits; a placebo that failed would invalidate the specification. Rank agreement is Spearman’s ρ against play-time-weighted raw win rates. Under W2 the ranking is preserved but effect sizes roughly halve (mean absolute difference 2.08 points; Squirrel Girl moves from -16.79 to -9.02), so the *ordering* should be treated as the result and magnitudes as an upper bound.

Table 5: Team-up synergies, largest and smallest

Team-up	Premium	Std. error
SQUIRREL MISSILE	8.66	(0.965)
BLADE OF KHONSHU	8.30	(0.851)
MR. & MRS. X	6.95	(0.472)
FLAWLESS DESIGN	6.82	(0.768)
KUMIHO PALM	6.61	(0.498)
STAR BLOSSOM	4.27	(0.503)
<i>... 90 team-ups omitted ...</i>		
COMPREHENSIVE DEFENSE	-1.80	(0.741)
STORMING IGNITION	-1.80	(1.057)
DIMENSIONAL SHORTCUT	-2.09	(0.666)
BINDING TIES	-2.24	(0.627)

The premium a pair earns *beyond* what its two heroes contribute individually, in percentage points, under W2 attribution. Median absolute effect across all 100 team-ups is 2.08 points. Squirrel Missile is the largest while Squirrel Girl is the weakest Damage hero individually, which is the separation of solo value from pair synergy the specification is built to make.

Limitations.

1. *Magnitudes depend on the attribution rule; the ranking is far more stable.* Across W0, W2 and W1 on an identical sample the within-role standard deviation runs 2.60, 4.27 and 6.38 points while rank agreement with W0 stays at $\rho = 0.937$ and 0.891 (Table 2). W0 is reported because it is the only rule fixed before any outcome information exists, but it is an intention-to-treat quantity and is diluted by players abandoning a hero mid-match. Read the ordering as the firmer result.
2. *These are adjusted associations, not causal effects.* Hero choice is not random. Pre-match rank score controls for general player skill but not for skill on the specific hero, and a one-trick is not a random player assigned that hero. Per-player hero playtime would measure this confounder directly; it was costed and declined, so it is bounded rather than eliminated.
3. *Intervals are analytic, not bootstrapped.* Matches are not independent draws — each contains twelve players and so belongs to twelve overlapping player clusters. The specified player-level cluster bootstrap would require roughly 656 hours at this scale under the current implementation, so HC1 intervals are reported instead. They ignore cross-classified dependence and are therefore *too narrow*; point estimates are unaffected.
4. *The sample is rank-selected.* Profile privacy rises from 0% below 4,000 rank score to 70% above 5,000, and the crawl can only expand through public players, so high-elo matches are structurally under-sampled. Further crawling does not fix this.
5. *Four team-ups are absent.* Hero id 1057 is base “Deadpool”, but the roster records only his three role variants, so 1057 is played zero times. The four team-ups defined against it are structurally empty and were dropped rather than mapped onto the variants, which would be an unverifiable modelling assumption.